Milestone 1

ECE 4950

Group 18

9/18/2026

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Executive Summary

Milestone 1 required our group to use MATLAB and Simulink with an Arduino controller, a potentiometer, and a servo motor to complete three tests involving sensor input and actuator control. The first test demonstrated the ability to receive an analog signal from the potentiometer and display the measured values as both numerical data and a plot in Simulink. The second test demonstrated our ability to turn the servo motor directly through commands issued in Simulink. The third test combined the sensor and actuator by using the potentiometer value to control the servo motor. When the potentiometer signal reached a specified threshold, the system changed the operating state of the servo motor. The threshold was adjustable so that the response of the system could be modified as needed.

In addition to the sensor and actuator testing, our group designed a small three-dimensional object for rapid prototyping. The part included our team number, Group 18, either embossed or debossed into the design and was designed to fit within the parameters: 4”x2”x1” (LxWxD). SolidWorks was used to create a three-dimensional model. Together, these activities introduced the group to sensor data acquisition, real time control using MATLAB and Simulink, Arduino hardware interfacing, and three-dimensional modeling for engineering applications.



Materials and Methods of the Sensor/Actuator Demonstration

Equipment and Software

Item	Manufacturer/ Model	Version/ Rating	Purpose
Software	MATLAB/ Simulink	2026a	Modeling data display and control
Micro Controller	Arduino Mega	2560	Sensor and actuator interface
Potentiometer	B103		Analog input sensor
Servo motor	MG92B		Actuator

Hardware Configuration

To begin Lab 1, the Arduino Mega was connected to the potentiometer and servo motor using jumper wires and a breadboard, as shown in Figure 1. The potentiometer was connected to the 3.3 V supply, analog input A4, and ground. The servo motor was connected to the 5.5 V supply, digital PWM pin D2, and ground. All required power and ground connections were provided by the Arduino Mega.

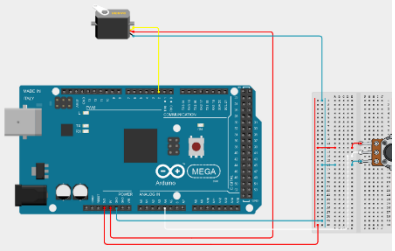


Figure 1. Circuit Diagram

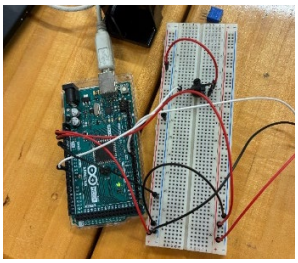


Figure 2. Physical Circuit

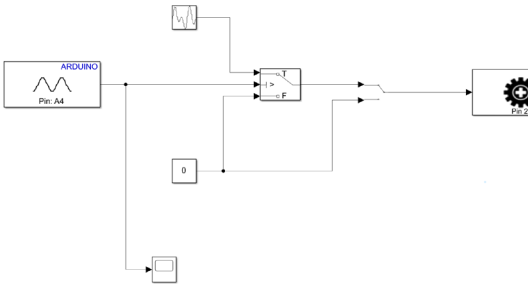


Figure 3. Simulink Model

Software Configuration and Procedure

The host computer was configured with MATLAB and Simulink version 2026a, along with the required Arduino support package. The Arduino was connected to the host using USB on port COM3. The Simulink model was then configured to communicate with the Arduino hardware, as seen in Figure 3. Simulink Model. The servo motor was driven by a sine wave with an amplitude of 90 to produce continuous movement. The potentiometer signal was connected to a switch that monitored whether the input exceeded a specified threshold, after which the motor was turned off. A second switch was placed after the main switching logic to provide manual on and off control of the servo motor. Finally, a scope block was connected to the potentiometer signal to monitor and display the measured values.

Results and Discussion

The Simulink model shown in Figure 3 was used to conduct the three required sensor and actuator tests described in the Milestone 1 assignment [3]. The Arduino was operated using the Monitor and Tune function in Simulink, allowing the system to receive live potentiometer measurements and control the servo motor in real time [1]. The individual experiments and their results are organized below.

Experiment 1: Potentiometer Signal Acquisition

Experiment 1 checked how well the system could read an analog signal from a potentiometer. To begin the potentiometer was linked to analog input A4 on the Arduino. The output from the potentiometer was watched in Simulink through a Scope block. When the potentiometer was turned to the right the reading went up. When it turned to the left the reading went down. These results showed that the Arduino and Simulink setup worked well. It was able to notice changes in where the potentiometer was and show the signal clearly.

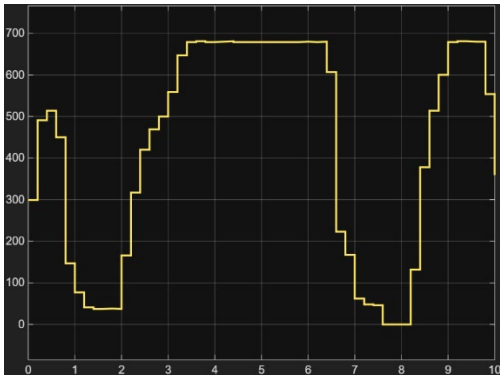


Figure 4. Potentiometer Readings Plot

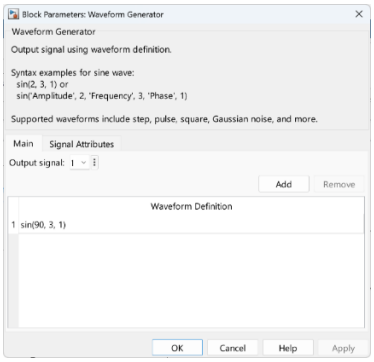


Figure 5. Sinewave Block Specifications

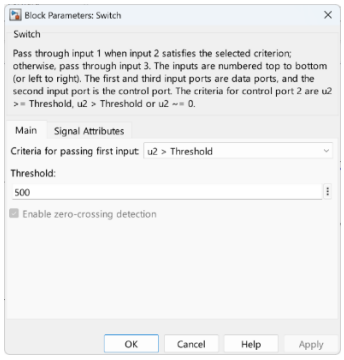


Figure 6. Threshold Block Value

The potentiometer appeared to output a minimum value of 0 and a max value of just under 700.

Experiment 2: Manual Servo Motor Control

The second experiment tested the ability to directly control the servo motor from Simulink. To achieve this, the Constant block in Simulink allowed the servo motor to receive a value from -90 to 90 to control the rotation speed. To change it to manual control, we utilized another switch to change the input to the servo to be the desired constant value.

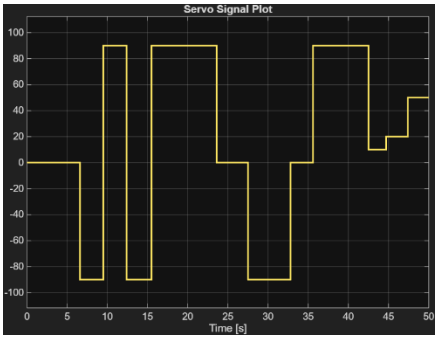


Figure 7. Manual Servo Control Plot

Figure 7 shows the servo command during the manual control test. Changing the selected input caused the command sent to the servo motor to change between the active and inactive states. The servo responded to these changes as expected, demonstrating that its operation could be controlled directly through Simulink.

Experiment 3: Potentiometer Threshold Control

The third experiment combined the potentiometer input with the servo motor control logic. A threshold value of 500 was selected in the Simulink switch as shown in **Figure 6**. While the potentiometer reading remained on one side of the selected threshold, the servo continued operating. As the potentiometer was rotated counterclockwise and the measured value reached approximately 500, the switching logic changed the command to the servo and stopped the motor.

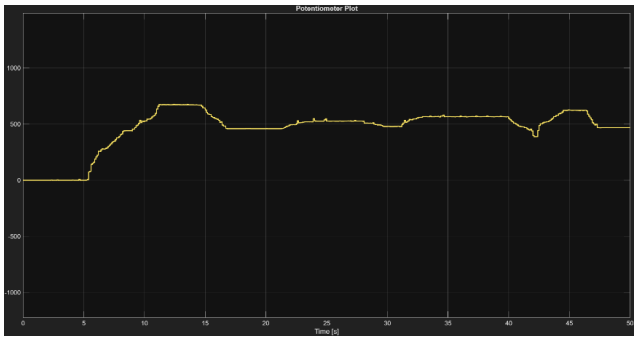


Figure 8. Potentiometer Waveform

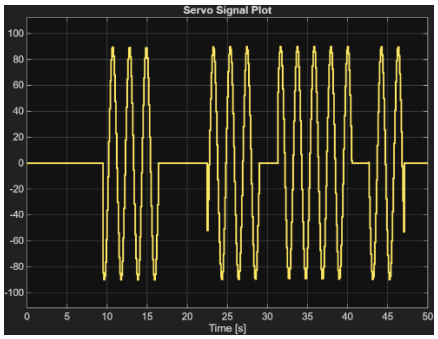


Figure 9. Servo Waveform

Figures 8 and 9 show the potentiometer input and corresponding signal during the threshold control test. Before the threshold was reached, the servo remained at rest. Once the potentiometer signal reached approximately 500, the switching logic changed states, and the servo motor began to move, using a sine wave with amplitude 90 as the input. While the servo was deactivated, further decreases in the potentiometer value did not reactivate the servo. The experiment demonstrated that a variable sensor input could automatically control an actuator through Simulink.

3D Designed Part

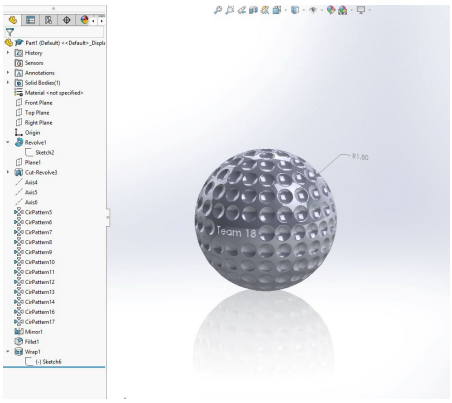


Figure10. Golf Ball Design

Figure 10 shows the three-dimensional model created for Milestone 1. The design consists of a golf ball shaped object with “Team 18” debossed into its surface. The model was created as an introduction to CAD design and rapid prototyping and was designed to remain within the dimensional requirements specified for the milestone.

Conclusion

Based on the results of the Milestone 1 experiments, the Arduino Mega and MATLAB/Simulink environment provide a useful platform for developing and testing basic robotic control systems. The experiments demonstrated that the Arduino could receive sensor input from a potentiometer and use that input to control a servo motor through Simulink. However, the potentiometer introduced some limitations. The originally provided potentiometer was difficult to adjust consistently, and the B103 potentiometer produced readings that were somewhat imprecise and did not always reach the same maximum value. These inconsistencies could make repeatable control more difficult in applications that require precise sensor measurements.

Three-dimensional modeling and printing may also support the mechanical development of the project. SolidWorks was used during Milestone 1 to design a golf ball shaped object with "Team 18" debossed on its surface while remaining within the required dimensions of approximately 4 in × 2 in × 1 in. Similar 3D printed parts could later be used to create structural components such as mounts for lights, cameras, motors, or other project hardware. Possible applications currently being considered include an arm for supporting the electromagnet and a base for the motor used to rotate the provided plate.

This milestone also introduced the group to the software, hardware, and development processes that will be used throughout the project. In addition to gaining experience with Arduino hardware, MATLAB, Simulink, and three-dimensional modeling, the group established methods for collaboration by creating a group chat and a shared Google Drive for communicating, developing documents, and exchanging files.

The Arduino Mega also has limitations that should be considered for more advanced robotic applications. Its physical size may be larger than necessary for compact systems, and other microcontrollers may provide better performance or efficiency depending on the final design requirements. Simulink provides a convenient graphical environment for developing and testing control logic, but model based or embedded code generation can introduce additional software and processing overhead compared with directly programming the microcontroller in languages such as C. Despite these limitations, the Arduino Mega provides an appropriate platform for the current project because it is accessible, relatively easy to configure, and capable of supporting the sensor and actuator control required at this stage of development.

References

[1] MathWorks, "Getting Started with Arduino Hardware," MATLAB and Simulink Support Package for Arduino Hardware. [Online]. Available: https://www.mathworks.com/help/supportpkg/arduino/ref/getting-started-with-arduino-hardware.html. [Accessed: Sep. 17, 2026].

[2] MathWorks, "Servo Control," MATLAB and Simulink Support Package for Arduino Hardware. [Online]. Available: https://www.mathworks.com/help/supportpkg/arduino/ref/servo-control.html. [Accessed: Sep. 17, 2026].

[3] Clemson University, Holcombe Department of Electrical and Computer Engineering, "ECE 4950 Technical Milestone 1: Setting Up a Device to Receive and Transmit Data," course handout, Fall 2026.

ECE 4950 Project 1 – Research Report Rubric

Each group will create a report that will eventually become a section in the **"Research"** section of your final project website. Use the guidelines below to complete your report and add at the end of your report.

Group Number (Group Name optional):

Group Member Last Names:_____

Score	Pts		
	15	**General Format - Professional Looking Document/Preparation (whole document)** a) Fonts, margins (11pt, times new roman, single spaced. 1" margins on all sides). b) Spelling and grammar are correct c) Layout of pictures – all figures need numbers and captions and must be referenced in the text d) Follows the page limitations below. e) References. Use IEEE reference format. f) This grading sheet is included as the final page.	**O3-SA1:1**
	15	**Page 1: Title, Group Name, Group Members, and Date** **Executive Summary** (~1/3 of the page) Provide a summary of the whole project. Use language that targets **a non-technical audience**. An important skill for an engineer is to communicate complex technical information to a general audience that may be involved in decision making, e.g. marketing. Important criteria: a) Can a non-technical audience (~ high-school degree) read this section and understand your goals, procedures, and conclusions? b) Use simple words and graphics to help explain	**O3-SA1:1**
	30	The next sections of the report follow the standard **laboratory report format**. **Page 2: Materials and Methods for the Sensor/Actuator Demonstration** (1 page) You are establishing the credibility and usefulness of your results by providing all the details so that someone else could repeat your experiment. As an example, MATLAB 2021a may behave differently than MATLAB 2021b – the software version information which would be required to reproduce your result should be included. This section should answer the following: a) What equipment is used, include software versions. **b)** How were the experiments conducted? How is the equipment connected and used? Describe the instrumentation, cables, connections, and experiments using diagrams and photos. You should have drawings (pin connection and connector part numbers) **Pages 3-4: Results and Discussion for the Experiments** (~1 page) Describe what you have done. Include plots (from MATLAB, not photos of the Target screen) for each of the three experiments and a brief discussion of how you interpret the result. Did you demonstrate (through your documentation) that the equipment has been configured and used correctly?	**O2-SA2:3** **O1-SA5:4**
	5	**Page 4-5: Conclusions and References** (~ 1 page) a) Based on this experiment, do you recommend this equipment for use in a robot control project? What are the possible limitations? Your results and observations should be the basis for your conclusions. (~1/2 page) b) What are the possible uses for the 3D Printer in your projects? (~1/4 page) c) What did you learn from the process of determining your final project design? **Page 6: This Grading Sheet**	**O3-SA1:1**
	5	**3D Designed Part** Grading based on: a) Does the design meet the requirements? b) Originality and creativity	**O7-SA1:1**
	10	**Demo Slide Show Presentation** (~3 slides) a) What did you do to complete this project? b) How did all the team members contribute? c) Photos of the designed 3D model	